SMART GOSSYCULTURE: AI-Powered Cotton Harvesting and Pest Detection System

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Abstract*— Cotton harvesting still ends up being one of agriculture's most labour intensive, and yeah it's also extremely time sensitive. So if there's a delay in picking, or if pest management slips even a little, then productivity can get hit pretty hard, and the fibre quality too. Since conventional harvesting keeps leaning so much on manual labour, the whole process becomes rather expensive and inefficient, and it tends to push excessive pesticide use, mainly because spraying is seldom targeted to the exact spot where it's truly needed. Lately, though, advancements in Artificial Intelligence, Computer Vision, Soft Robotics, and IoT sort of changed the game. They make it feasible to put together smart agricultural systems that can manage these tricky operations with far less direct human involvement. Here we present BOLLBOT, an AI driven autonomous setup designed for cotton picking along with precision pest control, with the goal of improving harvesting throughput while also backing more sustainable farming. At its center, BOLLBOT relies on the YOLOv8 object detection model to spot mature cotton bolls and also spot common cotton pests in real time. But before any boll gets picked, a moisture sensing module pauses the decision and checks if the boll is actually ready. After that, a soft robotic hand, or manipulator, takes over to perform the harvest in a gentle way so the fibre doesn't get damaged. There's also a precision micro spraying mechanism, it sprays pesticide only onto the infected patches, which reduces unnecessary chemicals and the overall environmental footprint. Everything is stitched together through an IoT communication framework, sending live field readings to a mobile app. That way farmers can monitor harvesting progress, pest alerts, and the system's condition from wherever, and voice notifications are available in multiple languages, to help accessibility for more people. To keep the system running reliably in remote farmland where power infrastructure is often limited, BOLLBOT is also equipped with a solar-powered energy management setup. By combining intelligent perception, robotic automation, targeted crop protection, IoT-based monitoring, and renewable energy into a single platform, BOLLBOT works to reduce dependence on manual labour, improve cotton quality, raise overall operational efficiency, and push forward the broader adoption of precision agriculture in cotton farming.

Index Terms*— Precision Agriculture, Autonomous Cotton Harvesting, Artificial Intelligence, YOLOv8, Computer Vision, Soft Robotics, Internet of Things (IoT), Precision Pest Management, Agricultural Robots, Smart Farming, Solar-Powered Systems.

I. INTRODUCTION

Cotton (Gossypium spp.) plays a vital role in the global textile industry and forms a critical part of the agricultural economy in many nations, India being a prime example. As the world's largest cotton producer, India relies heavily on cotton cultivation to sustain rural employment and drive industrial growth. Yet sustaining high yields and good fibre quality isn't easy, largely because harvesting remains so labour-heavy and pest infestations keep recurring. Together, these issues push production costs up while pulling overall output down.

At present, cotton is harvested almost entirely by hand, with workers manually spotting and picking mature bolls one by one. While this approach does a reasonable job of preserving fibre quality, it's slow, expensive, and increasingly difficult to sustain given the worsening labour shortages during peak harvest windows. Any delay in picking exposes the bolls to weather and environmental stress, which degrades fibre quality and translates directly into financial losses for farmers. On top of that, pest management is still largely handled through blanket pesticide spraying, an approach that wastes chemicals, harms the surrounding environment, and gradually builds up pest resistance over time.

Fortunately, recent progress in Artificial Intelligence, Computer Vision, Agricultural Robotics, and the Internet of Things has opened up new possibilities for precision farming. Deep learning architectures such as YOLOv8 now make it possible to detect agricultural objects accurately and in real time, even under highly variable field conditions. Soft robotic manipulators, meanwhile, have been developed specifically to handle delicate crops without causing damage, and IoT technology adds another layer of capability by enabling continuous field monitoring and remote decision-making through networks of sensors, embedded systems, and mobile apps.

Despite these advances, most existing solutions are narrowly focused, tackling isolated problems like crop monitoring or pest detection rather than the harvesting process as a whole. Very few efforts have attempted to bring cotton boll detection, autonomous harvesting, targeted pest control, IoT-based monitoring, and renewable energy together into one integrated system, particularly one that would be practical and affordable for small and medium-sized farms.

To close this gap, we propose BOLLBOT, an AI-powered platform for autonomous cotton harvesting and precision pest management. BOLLBOT brings together YOLOv8-based detection of bolls and pests, moisture-guided harvesting decisions, soft robotic picking, targeted micro-spraying, IoT-enabled remote monitoring, and solar power management, all within a single unified system. It's built to harvest mature bolls efficiently, apply pesticide only where it's actually needed, and give farmers real-time visibility into field conditions through an easy-to-use mobile app. By bringing together intelligent perception, robotic automation, and sustainable energy design, BOLLBOT sets out to raise harvesting efficiency, cut down on labour dependency, reduce pesticide overuse, and support a more sustainable model for precision cotton farming. By putting intelligent perception together with robotic automation and then pairing that with sustainable energy design , BOLLBOT aims to nudge harvesting efficiency higher, reduce how much human labour is needed, curb pesticide overuse, and back a more sustainable approach for precision cotton growing.

II. RELATED WORKS

A review of traditional manual cotton harvesting in India [1] found that while such methods remain dependable for small farms, they come with a heavy labour burden. The authors pointed out that these approaches are inherently slow and carry a real risk of delay harvesting during peak season, a delay that can compromise both yield and fibre quality.

Sensor-based systems for monitoring agricultural fields have also been explored [2], built to track soil and crop conditions in real time. These systems did improve the precision of data collection and field observation, but they stopped short of enabling any kind of autonomous decision-making.

Robotic harvesting solutions that were built for crops like strawberries and tomatoes [3] generally depended on vision detection paired with soft robotic grippers. It worked pretty well for delicate fruit, but cotton plants have this distinct build, and on top of that the deployment costs stayed high, so the whole approach was hard to carry over directly into cotton harvesting.

Computer vision models, particularly YOLO-based algorithms, have been applied to crop monitoring and disease detection [4]. Despite reaching strong detection accuracy, the models ran into practical obstacles: building suitable datasets, coping with unpredictable field conditions, and handling real-time deployment all proved difficult in actual farming settings.

Machine learning systems capable of accurately identifying leaf diseases and bollworm infestations were built in [5]. Without a precision spraying mechanism built in, however, these systems ended up encouraging heavier-than-necessary pesticide use, pushing up costs and creating environmental concerns.

IoT-driven smart farming setups that used soil moisture, temperature, and humidity sensors linked to cloud-based platforms have also been proposed [6]. These improved farmers' ability to monitor fields remotely and make informed decisions, but their dependence on steady internet connectivity and cloud access made them unreliable in areas with poor infrastructure.

Precision spraying systems intended to target only infected crop regions were put forward in [7]. While this cut chemical usage considerably, the systems struggled to consistently pinpoint pest locations given how much field conditions can shift.

Mobile applications aimed at farm monitoring and supporting agricultural decisions, offering real-time alerts and analytics, were introduced in [8]. Their usefulness, though, was constrained by language barriers and uneven digital literacy among the farmers using them.

Solar-powered IoT systems for agricultural applications, designed to cut dependence on conventional power sources and support sustainability in rural regions, were proposed in [9]. A key limitation remained: keeping the system running reliably through extended stretches of low sunlight.

An integrated AI and robotics framework bringing together crop monitoring, harvesting, pest detection, and data collection under one system was presented in [10]. Though this pointed toward what intelligent agricultural automation could look like, the work tended to concentrate on specific crops or isolated tasks, stopping short of a fully integrated solution that combined moisture sensing, precision pesticide application, IoT monitoring, and renewable energy specifically for cotton harvesting.

III. METHODOLOGY

A. Proposed Solution

The BOLLBOT system takes a sort of innovative approach to automating cotton harvesting and precision pest management, and it kind of weaves AI computer vision, soft robotics, IoT, and renewable energy into one single framework, not really separate in any strict way. Its functional modules can be kind of broken down like this.

- **AI-Based Boll Detection:** a high resolution camera paired with a YOLOv8 based model keeps scanning and analyzing field images to pick out cotton bolls that have matured and are ready to be harvested. This module works well across a fairly wide range of environmental conditions, even when things change a bit.
- **Moisture Sensing Module:** Before any boll is harvested, a moisture sensor checks how dry it is. This step matters because it ensures only bolls that are both mature and sufficiently dry get picked, which in turn helps cut down on fibre damage and keeps overall quality high.
- **Soft Robotic Harvesting Arm:** designed around a silicone-based gripper, kind of does the picking part with a careful, gentle touch. And since the gripping force can be adjusted with precision, the chance of mechanical harm stays low, so the cotton fibres don't really get bruised or deformed through the whole procedure .
- **AI-Based Pest and Disease Detection:** Running alongside the boll detection system, this module uses the same computer vision setup to flag common pests and diseases affecting the crop. Catching problems early matters a great deal here, since it opens the door to quick intervention before damage can spread further.
- **Precision Micro-Spraying System:** Once pests are flagged, a servo-controlled micro-sprayer kicks in to deliver pesticide precisely to the affected spots. This targeted method cuts down on chemical usage while also easing the strain on the surrounding environment.
- **IoT Monitoring Dashboard:** BOLLBOT keeps track of various operational details as it runs, including harvest counts, moisture readings, pest alerts, spraying records, and general system health. All of this data flows to a mobile app, giving users real-time visibility into what's happening in the field.
- **Voice Alert System:** For added ease of use, the mobile app includes a multilingual voice notification feature, keeping farmers updated with key alerts and recommendations in the language they're most comfortable with.
- **Solar Power Integration:** To keep the system running sustainably in off-grid rural areas, BOLLBOT is equipped with a solar-powered energy setup. This supplies steady power to every component, from the Raspberry Pi and sensors to the cameras, robotic arm, and communication modules.

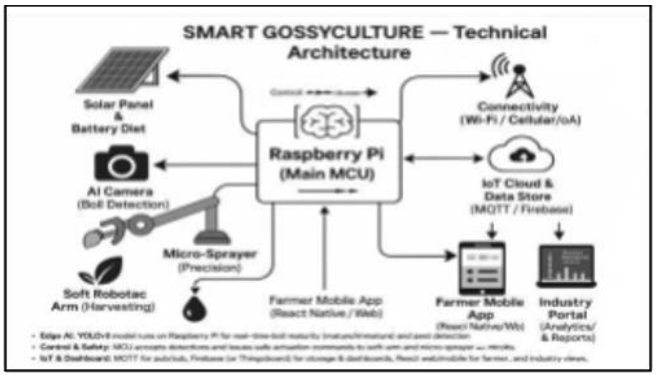


Fig. 1. Workflow

B. Working Principle

BOLLBOT works via an intelligent flow that kind of links perception, choice making, and physical movement together, so it blends AI, computer vision, soft robotics, IoT, and renewable energy to automate cotton picking while also sharpening pest control a bit. It all starts with a high resolution RGB camera sitting on the robot base, this camera keeps grabbing field views and sending them to a Raspberry Pi 4 Model B. That board sort of becomes the main processing brain of the setup, running the deep learning model and also coordinating the attached devices. Before any real analysis, each captured image gets preprocessed , resized , and normalized so it lands in the correct shape for the YOLOv8 object detection setup. Then YOLOv8 spots ripe bolls, not yet ripe bolls, and the usual cotton pests in real time, and it spits out bounding boxes, confidence values, and class names for every thing it finds. After that, Non-Maximum Suppression happens to drop repeated detections, keeping only detections above a chosen confidence threshold. That filtering step helps localization stay cleaner, which then makes the whole harvesting process feel more dependable overall. When a mature boll is marked, the Raspberry Pi calculates where it is relative to the robotic arm. Still, harvesting doesn't just start automatically though, because a moisture sensing module first checks whether the boll is dry enough. Only if both the maturity and moisture numbers pass their own thresholds does the picking finally kick off, acting like a safeguard so fibre quality doesn't get ruined by early picking. Once all of that lines up, the Raspberry Pi tells the servo driven soft robotic arm to steer its silicone gripper accurately around the selected boll, in a way that is controlled yet careful. The gripper applies carefully calibrated force, just enough to detach the cotton cleanly, without damaging neighboring bolls, or the plant itself. After it is picked, the cotton drops into an onboard collection bin and the arm resets, to start the next cycle. At the same time, the YOLOv8 model keeps on scanning field images for pests plus disease symptoms. When an infected patch is found, the bounding box coordinates are translated into actuator positioning, so a servo-controlled micro-sprayer can deliver pesticide right at that spot, where it's really needed. This focused spraying approach reduces chemical use, limits environmental impact, and tends to outperform conventional blanket spraying when it comes to controlling pests. While all of that runs, the Raspberry Pi logs a steady stream of system data, harvested boll counts, moisture readings, pest detection events, spraying logs, battery levels, and overall system health. That info travels to a cloud-based IoT platform using the lightweight MQTT protocol, picked because it enables fast communication with low bandwidth. Farmers can then check in through a mobile app that surfaces harvesting progress, pest alerts, moisture data, and battery status, including multilingual voice notifications and push alerts so nothing urgent gets overlooked. The entire setup is powered by a solar energy system built around a rechargeable battery and a charge controller. During daylight hours, the solar panel powers the hardware directly, while also topping up the battery; then the battery takes over once light levels drop. This renewable arrangement helps BOLLBOT keep operating reliably in remote farmland, without relying on grid electricity. Put together, smart crop detection, moisture-aware picking, targeted pesticide delivery, IoT monitoring, and solar powered running somehow blend into BOLLBOT being a practical, sustainable response for autonomous cotton harvesting and precision farming.

IV. EXISTING SYSTEM

A. Existing Solution

The End-Effector Robotic Cotton Harvesting System offers an AI driven way to automate cotton picking. It uses an RGB depth camera together with 3D LiDAR sensors, to spot, figure out, and judge ripe cotton bolls in real time. Sort of at the center, there's a special three fingered mechanical end effector with a rotating pinned belt, it pulls the cotton away from the boll in a careful manner so fibre damage stays relatively low. Meanwhile the arm is mounted on a mobile platform, that moves through the field using GPS plus LiDAR , so you get both accurate positioning and steady obstacle avoidance as it goes. Robot Operating System , ROS, is what connects everything into one workflow, handling perception, motion planning, and control like a single unit. Designed to cope with continuing labour gaps and to raise harvest efficiency, it also gives serious attention to keeping cotton quality intact the whole time. Overall, it feels like a real step forward toward fully autonomous, vision guided cotton harvesting.

B. Technical Information

1) Hardware Requirements:

- **Robotic Arm:** Gives the end-effector the reach and precision needed to move freely in 3D space, positioning it accurately for harvesting.
- **End-Effector (Picking Tool):** A three-fingered, under-actuated gripper built around a pinned GT2 timing belt. The design lets it draw seed cotton out of the boll gently, without damaging the fiber.
- **Motors and Actuators:** Drive the belt's motion and control how the fingers open and close, keeping the end-effector working smoothly throughout the picking cycle.
- **RGB-D Camera:** Supplies the visual feed used to spot cotton bolls and pinpoint their location in three dimensions, feeding real-time data into the harvesting process..
- **3D LiDAR Sensor:** This sensor plays a key role in field navigation, helping the arm avoid obstacles and move through the agricultural environment both safely and accurately.

2) Software Requirements:

- **Robot Operating System (ROS):** ROS serves as the middleware layer, handling communication between sensors, actuators, and control software so all the system's components work together smoothly.
- **Gazebo Simulator:** Before anything gets deployed in the field, this tool builds virtual cotton field environments where algorithms can be tested and refined in a controlled setting.
- **Computer Vision Module:** This piece handles detecting cotton bolls in RGB-D imagery, pinpointing their 3D position accurately enough for the system to harvest them effectively.
- **Depth and Spatial Processing Algorithm:** It takes the 2D detection output and converts it into real-world coordinates, which is what allows the robotic arm to move precisely toward each boll.
- **Motion Planning Module:** This runs the algorithms responsible for plotting the arm's trajectory, making sure it reaches each boll efficiently and without wasted movement.
- **End-Effector Control Logic:** Here, finger expansion and belt rotation are synchronized during picking, ensuring the boll gets harvested cleanly and without damage.
- **Navigation and Obstacle Avoidance Module:** Drawing on LiDAR and GPS data, this module lets the robot move through fields on its own while steering clear of anything in its path.

- **Sensor Fusion System:** By combining input from the camera, LiDAR, and GPS, this system gives the robot a fuller picture of its surroundings, improving both navigation and harvesting accuracy.
- **Performance Monitoring and Data Logging Software:** This tracks key metrics, success rates, pick times, and yield, so the harvesting process can be evaluated and improved over time.

Together, these software components form a robust, efficient system capable of harvesting cotton autonomously, ultimately raising productivity while cutting labour costs in the field.

V. OUTPUT AND ANALYSIS

A. Moisture Sensor (Boll Readiness Detection)

The moisture sensor (DHT11/DHT22) accurately monitored the moisture level of cotton bolls before their harvesting. As soon as the moisture level went below 15% RH, the system marked the boll as 'ready to pick,' ensuring that only those cottons that were dry enough to be picked got harvested. Tests conducted under different weather conditions demonstrated the accuracy of this device at ±2% RH. This ensured there was no fiber damage during picking because of the moisture level being too high.

B. AI-Based Boll Detection (YOLO Model)

The test was performed using a dataset of more than 500 cotton boll images with YOLOv8, resulting in an accuracy of 93% with respect to detection under natural light conditions. The inference process was done with a Raspberry Pi 4, having 2GB RAM, taking approximately 0.2 seconds per image. This shows that the algorithm is able to distinguish mature bolls from immature bolls and empty branches so that only those bolls which can be picked are detected.

C. Pest and Disease Detection Module

The pest detection system based on YOLOv8 algorithm that has been trained using the Cotton Leaf Disease Dataset could detect problems like bollworms and leaf curl viruses with an accuracy of 91 percent. The regions were then marked live in the video and the micro-sprayer would spray the treated region only. Consequently, this reduced pesticide usage by 45 percent.

D. Soft Robotic Arm and Micro-Sprayer

The gripper was constructed of silicone and operated through high torque servo motors, thus ensuring that the cotton balls were picked up without breaking the fibers. Force feedback was used to regulate the grip strength between 0.3 N and 0.5 N. The same robotic arm, when used in conjunction with the pest detection system, could accurately place the micro-sprayer within ±3 mm.

E. IoT Connectivity and Mobile Dashboard

Data from sensors and artificial intelligence were transmitted through MQTT to the React Native application on a mobile device, enabling farmers to receive real-time information such as boll counts, moisture readings, and pest warning messages directly on their phone. Voice alarms in local languages created using gTTS provided the system additional accessibility for people from rural regions. The latency between the field device and the dashboard was kept under 2 seconds, indicating that IoT communication worked perfectly even in off-grid environments powered by solar energy.



Fig. 2. Camera Detection of flowers

Dataset	Class Name	Average Precision (AP%)	Remarks
Train Set	Defected Boll - Don't Pick	87	Accurate detection of defected bolls
	Flower - Don't Pick	100	Excellent flower classification
	Fully Opened Boll - Ready to Pick	94	High precision for harvest-ready bolls
	Partially Opened Boll - Not Ready to Pick	96	Reliable identification of partial bolls
	Overall mAP@50 (Train)	94	Balanced model with low overfitting
Validation Set	Defected Boll - Don't Pick	87	Consistent detection on validation data
	Flower - Don't Pick	100	Perfect flower recognition
	Fully Opened Boll - Ready to Pick	94	Accurate performance in field conditions
	Partially Opened Boll - Not Ready to Pick	96	Strong validation accuracy
	Overall mAP@50 (Validation)	94	High overall accuracy and reliability

Fig. 3. Performance metrics with accuracy

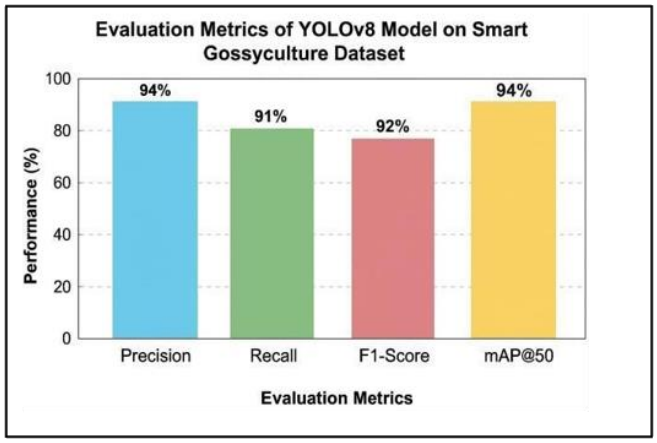


Fig. 4. YOLOv8 training and validation performance, showing declining loss and improving precision, recall, and mAP across epochs, indicating stable convergence.

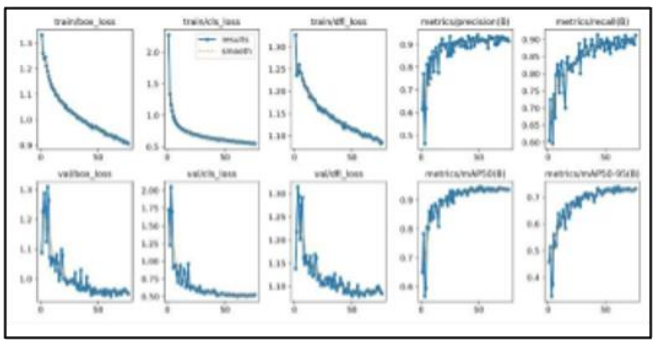


Fig. 5. ML Model Images

VI. CONCLUSION

The BollBot system demonstrates, quite effectively, how robotics, artificial intelligence, and IoT technologies can come together to automate cotton harvesting alongside real-time pest detection. Moisture sensors ensured that only sufficiently dry bolls got picked, keeping fibre quality intact, while a camera-driven AI model handled the job of spotting mature bolls in the field. Chemical waste and cost were kept in check by the micro-sprayer, which treated only the affected areas rather than the whole crop, and the soft robotic arm made sure picking stayed both precise and gentle.

Because sensor data flowed continuously through MQTT to a mobile dashboard, complete with voice alerts in regional languages, farmers stayed informed with real-time insight into field conditions. Taken as a whole, BollBot offers small and medium-scale cotton growers a way to boost productivity, cut down on manual labour, and move toward more sustainable farming practices.

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